

CM 7028 Digital Media Production



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L2

นน 7028 การผลิตสื่อดิจิทัล

INSTRUCTOR

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- (Please Email/Facebook me first to make sure that I will be there.)

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Course Syllabus and Lectures

All course syllabus and lectures can be downloaded online via the following links.

- Course Syllabus
 - <http://gscm.nida.ac.th/~chutisant/CM7028/CS.pdf>
- Lectures
 - <http://gscm.nida.ac.th/~chutisant/CM7028/L1.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM7028/L2.pdf>

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Overview

- Digital Media by New Technologies
- Augmented Reality
- Computer Vision

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Digital Media by New Technologies

Digital Media

Digital media are any media that are encoded in a machine-readable format. [*Technology Brief*. University of Guelph.]

1. Digital Imagery / Digital Video
2. Web Pages / Websites
3. Digital Audio, e.g. mp3s
4. e-books
5. Social Media



www.linkedin.com

Mobile Phone As Home Computer

- Mobile Phone as PC
- Mobile phones are the new computer
- Designing website/social media for mobile phones is undoubtedly essential



voices.suntimes.com

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Creative-Digital Economy

- Placement: Its where young (and old) people are
- Price to invest: Minimal
- But..
 - Easy to build innovatively
 - Create HUGE profits



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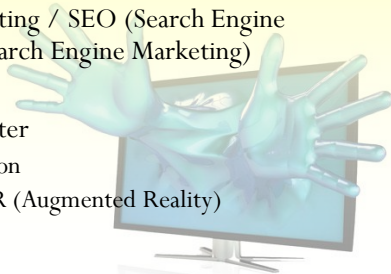
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<http://www.dailynews.co.th/>

The Future of Digital Media

Corporate Image and Reputation Management using Technologies of the Future

- Big Data / Cloud Computing / SEO (Search Engine Optimization) – SEM (Search Engine Marketing)
- Mobile-Oriented
- Reaching-Out-of-Computer
 - 3D-Display / 3D Television
 - MR (Mixed Reality) / AR (Augmented Reality)



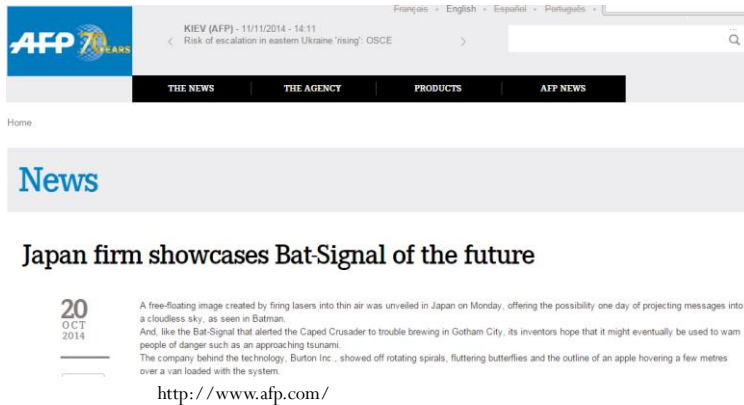
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www.3dsg.com.au

3D Display / 3D Television

3D Display / 3D Television



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3D Display / 3D Television

A free-floating image created by firing lasers into thin air



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<http://www.afp.com/>

3D Display / 3D Television



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<http://www.afp.com/>

3D Display / 3D Television

- Miraikan is located in Odaiba, Tokyo



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www.whitehouse.gov



www.japan-guide.com

New Viewpoint Analysis

Prof. Takeo Kanade,
Carnegie Mellon University (CMU),
United States

Super Bowl

- “Eye Vision”
- Involves shooting multiple video images of a dynamic event, such as a football game, from multiple cameras placed at different angles
- The video streams from these cameras are combined by computer and the resulting images reach viewers in a format that will make them feel as if they are flying through the scenes they see.



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Eye Vision from Prof. Takeo
Kanade, CMU

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The Matrix



www.imdb.com

- “Virtualized Reality”
- People will be able to customize the perspective from which they watch, for example, from that of a particular object.

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New Viewpoint Analysis

Augmented Reality

What is Augmented Reality (AR)?



- A combination of a real scene viewed by a user and a virtual scene generated by a computer that augments the scene with additional information.
- Generates a composite view for the user.

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The Ultimate Goal of AR

- Create a system such that no user CANNOT tell the difference between the real world and the virtual augmentation of it.

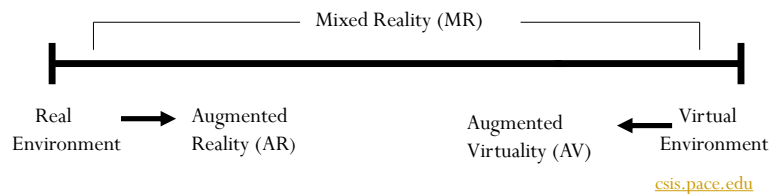


AR in Commerce Geek.com

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Miligram's Reality-Virtuality Continuum



Miligram coined the term "Augmented Virtuality" to identify systems which are mostly synthetic with some real world imagery added such as texture mapping video onto virtual objects.

Augmented Reality is closest to the real world because mainly a user is perceiving the real world with just a little computer generated data.

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Combining the Real and Virtual Worlds

We need:

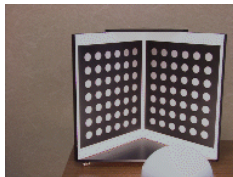
- Precise models
- Locations and optical properties of the viewer (or camera) and the display
- Calibration of all devices
- To combine all local coordinate systems centered on the devices and the objects in the scene in a global coordinate system

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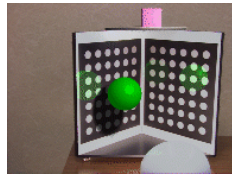
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Combining the Real and Virtual Worlds (cont)

- Register models of all 3D objects of interest with their counterparts in the scene
- Track the objects over time when the user moves and interacts with the scene



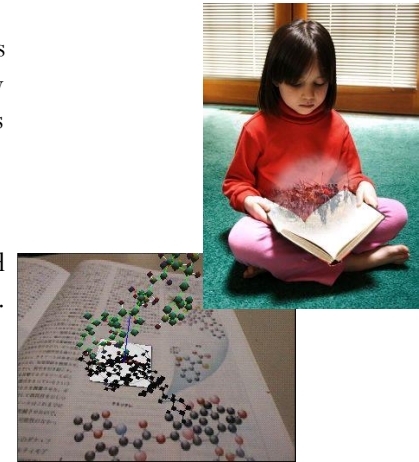
real world



real world with virtual objects and inter-reflections and virtual shading

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This image depicts an artist's conception of a mixed reality enhanced book. The reader is able to read the actual book, but simultaneously see three dimensional computer graphics co-located with, and in registration with the book.



www.sonycsi.co.jp

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Realistic Merging

Requires:

- Objects to behave in physically plausible manners when manipulated
- Occlusion
- Collision detection
- Shadows

****All of this requires a very detailed description of the physical scene**

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Research Activities

- Develop methods to register the two distinct sets of images and keep them registered in real-time
 - New work in this area has started to use Computer Vision (CV) techniques
- Develop new display technologies for merging the two images

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Performance Issues

Augmented Reality systems are expected:

- To run in real-time so that the user can move around freely in the environment
- Show a properly rendered augmented image

Therefore, two performance criteria are placed on the system:

- Update rate for generating the augmenting image
- Accuracy of the registration of the real and virtual image

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Applications

- Medical
- Military Training
- Hazard Detection
- Engineering Design
- Robotics and Telerobotics
- Training
- Sports
- Tourism
- Video conferencing

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Applications (cont.)

- Entertainment, Computer Games
- Manufacturing, Maintenance, and Repair
- Consumer Design
- Audio
- Education
- Architecture
- Human welfare

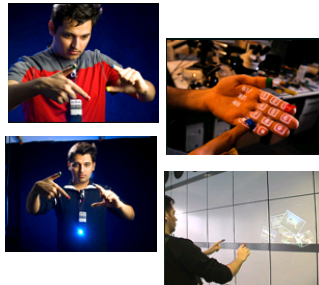
Recent Augmented Reality Applications

Sixth Sense

Sixth Sense

- A wearable gestural interface that augments the physical world around us with digital information
- Let us use natural hand gestures to interact with that information
- The SixthSense prototype is comprised of a pocket projector, a mirror and a camera.

Sixth Sense in 21st Century



Dr. Pattie Maes,
Massachusetts Institute of Technology
(MIT), United States

Broadcast in TED

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TED
TALKS

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Detail of Sixth Sense

- Sixth Sense allows people to use the internet without a screen or a keyboard
- It can turn any surface into a touch screen
- It is made with store-bought components
- *“The software program processes the video stream data captured by the camera and tracks the locations of the colored markers [that you wear on your fingers] using simple computer-vision techniques”..from MIT Media Lab*

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Augmented Reality Guitar

Augmented Reality Guitar



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Keio University, Japan

Broadcast in Discovery Channel

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Broadcast in Discovery Channel



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MR for Military Application

Mixed Reality for Military Application



Dr. Adrian Cheok,
National University of Singapore (NUS)

Broadcast in CNN



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Augmented Reality Magic

Augmented Reality Magic

- Now that an ever-accelerating cascade of eye-popping visual technology such as AR has threatened to steal some of the magic dust from old-fashioned magicians
- Along comes a pasteboard prestidigitator who folds augmented reality into the magic world



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Augmented Reality Magic (card trick)

- Blend of technology and illusion creates the magic of tomorrow, today



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Integration leads to Innovation

Information Technology
Research (Eye Tracking)



Comm. Arts Research
(Advertising Strategy)



Research
Innovation

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Graduate School of Communication Arts and Management **Innovation**
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Stanford University
STANFORD UNIVERSITY

Department of
Communication

Prospective Students
Student Resources
People
Research
Journalism

Communication and Technology

The Stanford Department of Communication has been a pioneer in studying the relationship between people and digital media for decades. In the early nineties, Clifford Nass and his graduate students were among the first in the world to empirically examine constructs such as agency and anthropomorphism. Soon thereafter Byron Reeves and Clifford Nass published their landmark book, *The Media Equation*, which set the stage for a new research paradigm based on the notion that the brain has not evolved to differentiate mediated experiences from actual ones. Indeed, one of the fastest growing divisions of the International Communication Association, Communication and Technology, is substantially based on the work of Stanford scholars.

Currently, the area focuses on understanding the ways in which digital media is changing the way we communicate and the ways in which we are being changed by it. The area also draws on collaborations with the MediaX. Moreover, the area is also interested in "social networking" research.

Faculty

DATA-DRIVEN

A new lineup of courses that focus on DATA JOURNALISM.

STANFORD COMPUTATIONAL JOURNALISM LAB

Coming soon

Our new lab will focus on developing tools that will help lower the costs for journalists to discover stories.

Stanford_Journalism

<http://comm.stanford.edu/technology/>

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Oxford Internet Institute
Understanding life online

DPhil in Information, Communication and the Social Sciences

The DPhil in Information, Communication and the Social Sciences (doctoral) programme provides an opportunity for highly qualified students to undertake cutting edge Internet-related research. We are looking for academically excellent candidates who display the potential and enthusiasm necessary to perform research that will make a difference -- to ask important questions and to adopt innovative methodologies and approaches for exploring those questions. Many OII doctoral students are pursuing research that will shape the development of digital networked spaces and those whose lives are affected by it.

OII Research Fellow Kathryn Eccles has been appointed 'Digital Humanities Champion' for the University of Oxford. She will help develop the cross-University Digital Humanities strategy.

<http://www.oii.ox.ac.uk/>

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KEIO MEDIA DESIGN.

慶應義塾大学大学院メディアデザイン研究科
GRADUATE SCHOOL OF MEDIA DESIGN, KEIO UNIVERSITY

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Keio University

General Info

Digital media is changing from conventional mass media to personal media that uses media infrastructure for community building. Examples include the convergence of broadcasting and communications and the emergence of social networking services. Content adds value to media, and design enhances all aspects of life. These have become important engines in enriching the human heart and mind. For media, content, and design to take root in society, it is crucial to understand how they function as industries, to understand their business and management models, and to include policies about intellectual property & describe in "Our philosophy of education," we train "media innovators," defined as people who



KEIO MEDIA DESIGN

慶應義塾大学大学院メディアデザイン研究科
GRADUATE SCHOOL OF MEDIA DESIGN, KEIO UNIVERSITY

<http://www.kmd.keio.ac.jp/>

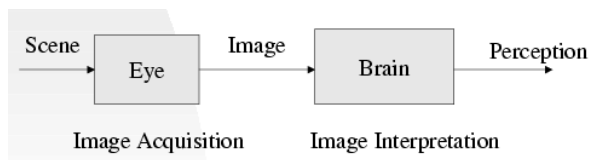
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Graduate School of Media Design

Introduction to Computer Vision

Vision

- Vision is the process of discovering what is present in the world and where it is by looking.



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Computer Vision

- “**Computer vision** is the science and technology of machines that see. As a scientific discipline, computer vision is concerned with the theory behind artificial systems that extract information from images. The image data can take many forms, such as video sequences, views from multiple cameras, or multi-dimensional data from a medical scanner.”

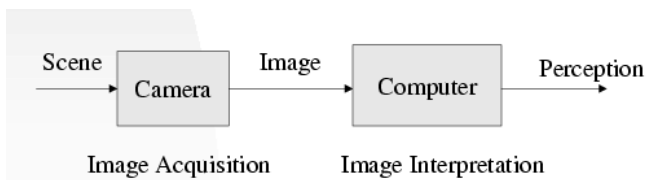
By Dana H. Ballard, Computer Vision.

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Computer Vision

- Computer Vision is the study of analysis of pictures and videos in order to achieve results similar to those as by men.



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Introduction of Computer Vision

- Digital Image Processing (IP)
 - Since 1960's
 - 1970's : Recognized as a technology for practical use.
 - Most Important Issues : Image Transformation
 - Noise/Blur Compensation, Distortion Correction

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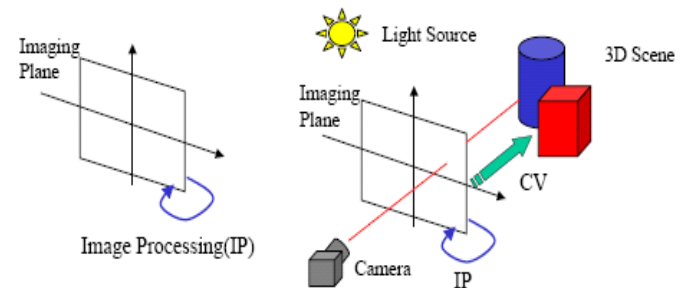
Introduction of Computer Vision

- Computer Vision (CV)
 - Emerged from AI (Artificial Intelligence) Research
 - Computational Vision proposed by Marr
 - Shape from X(X:shading, texture, contour, motion...etc.)
 - 3D Scene Understanding from 2D images
 - Robot Vision: Vision functionality for Robots
 - Media Processing, Human Interface

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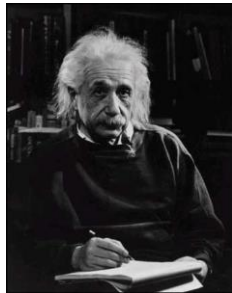
Image Processing / Computer Vision



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Computer Vision



What we see

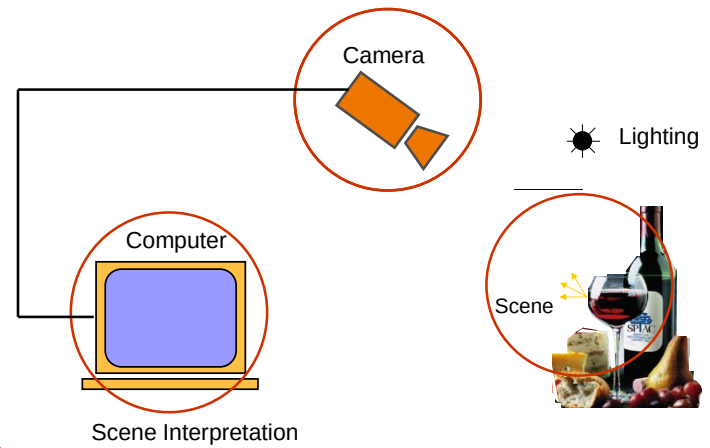
0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a computer sees

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Components of a Computer Vision System



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Example

Finding People in images

Problem 1: Given an image I

Question: Does I contain an image of a person?

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“Yes” Instances



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“No” Instances



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Thank you

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